

AI Readiness Scale (AIRS)

Development and Validation of a Psychometric Instrument for Assessing Enterprise AI Adoption

DOCTOR OF BUSINESS ADMINISTRATION

N = 523 · OCTOBER–NOVEMBER 2025



The Adoption-Value Gap

88%

AI Tool Usage

Among knowledge workers in 2025, up from
50% in 2023

5%

Value Realized

Organizations reporting measurable value from
AI implementations

90–95%

Pilots Stalled

Generative AI pilot programs that never
progress beyond initial trials

Organizations invest heavily in AI tools yet consistently fail to realize meaningful returns. This adoption-value gap — high investment, low return — is the central problem motivating this research.



Three Interconnected Research Gaps

Theoretical Gap

UTAUT2 was validated for consumer mobile technologies — not AI. Whether its predictor structure holds for autonomous, probabilistic AI tools remained empirically untested.

Measurement Gap

No validated psychometric instrument existed for AI-specific adoption readiness. Organizations lacked a diagnostic tool to assess workforce readiness for enterprise AI deployment.

Practice Gap

Without systematic measurement, AI implementation strategies remained generic rather than responsive to each workforce's particular readiness profile.

The Solution: AIRS

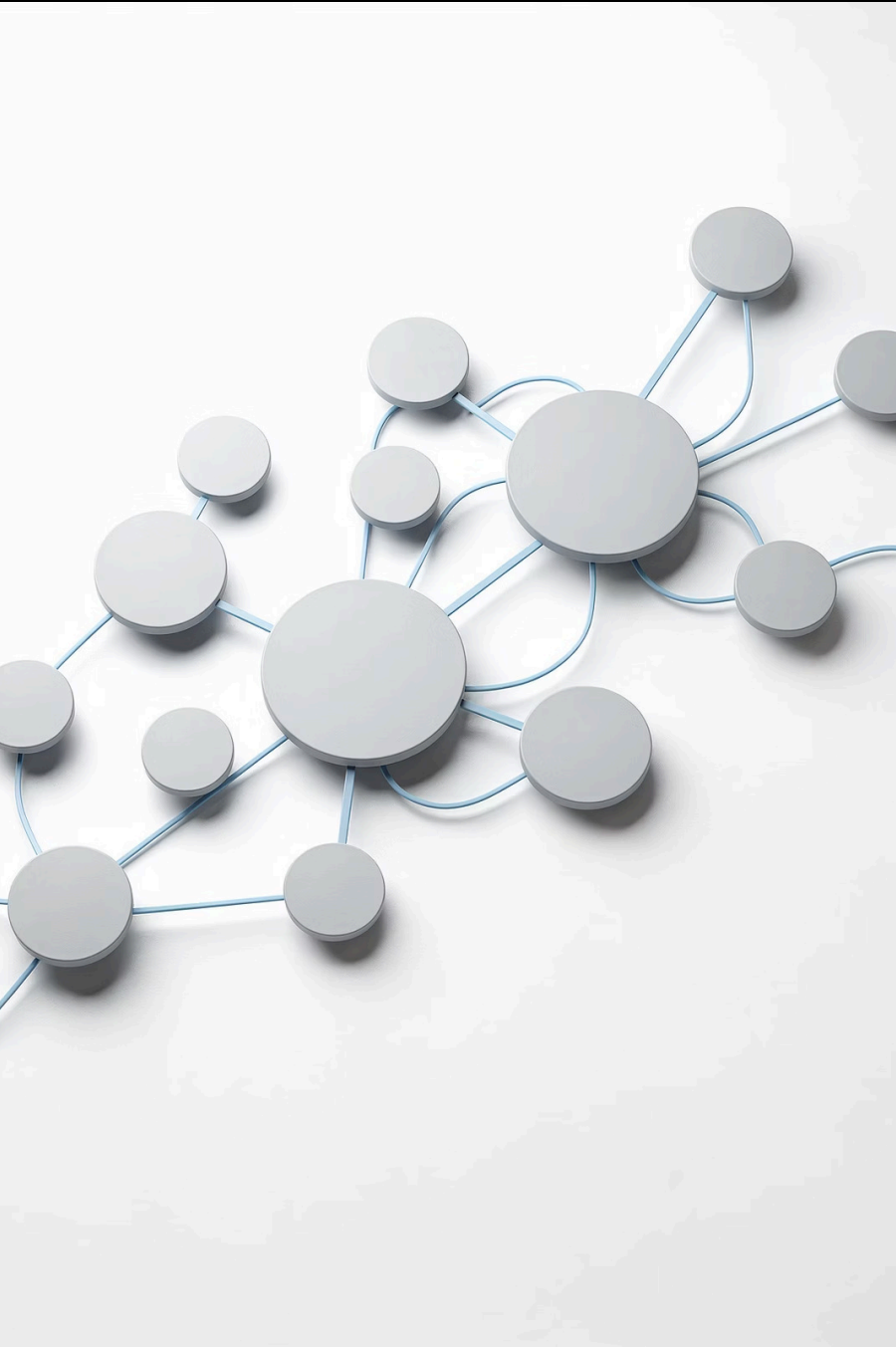
What Was Built

The AI Readiness Scale (AIRS) — a 16-item psychometric instrument extending UTAUT2 with an AI Trust construct.

8 factors × 2 items each, plus 4 Behavioral Intention outcome items.

What It Delivers

- Theoretical account of AI adoption drivers
- Practical diagnostic tool for organizational assessment
- Evidence-based framework for targeted interventions
- Validated across N = 523 U.S. students and professionals



CHAPTER 1

Theoretical Framework

The study builds on UTAUT2 – the most comprehensive validated model of technology acceptance – and extends it with a novel AI Trust construct to address the unique characteristics of artificial intelligence.

UTAUT2: Seven Core Constructs

Performance Expectancy

Perceived improvement in job performance from using the technology

Effort Expectancy

Perceived ease of learning and using the technology

Social Influence

Degree to which important others believe one should use the technology

Facilitating Conditions

Perception of available resources and support infrastructure

Hedonic Motivation

Pleasure or enjoyment derived from using the technology

Price Value

Perceived balance between benefits received and monetary cost

Habit

Extent to which technology use has become automatic

The 8th Construct: AI Trust

AI tools differ fundamentally from conventional technologies through **autonomous reasoning, opaque decision processes, and evolving behaviors**.

AI Trust captures the degree to which users believe AI systems will provide accurate, reliable information and operate in a trustworthy manner — a unique barrier not adequately captured by existing UTAUT2 constructs.

Trust Items

TR1: I trust AI tools to provide reliable information.

TR2: I trust the AI tools that are available to me.

Research Questions & Hypotheses

01

Primary RQ

To what extent does UTAUT2, extended with AI Trust, explain Behavioral Intention to adopt AI tools among U.S. students and professionals?

02

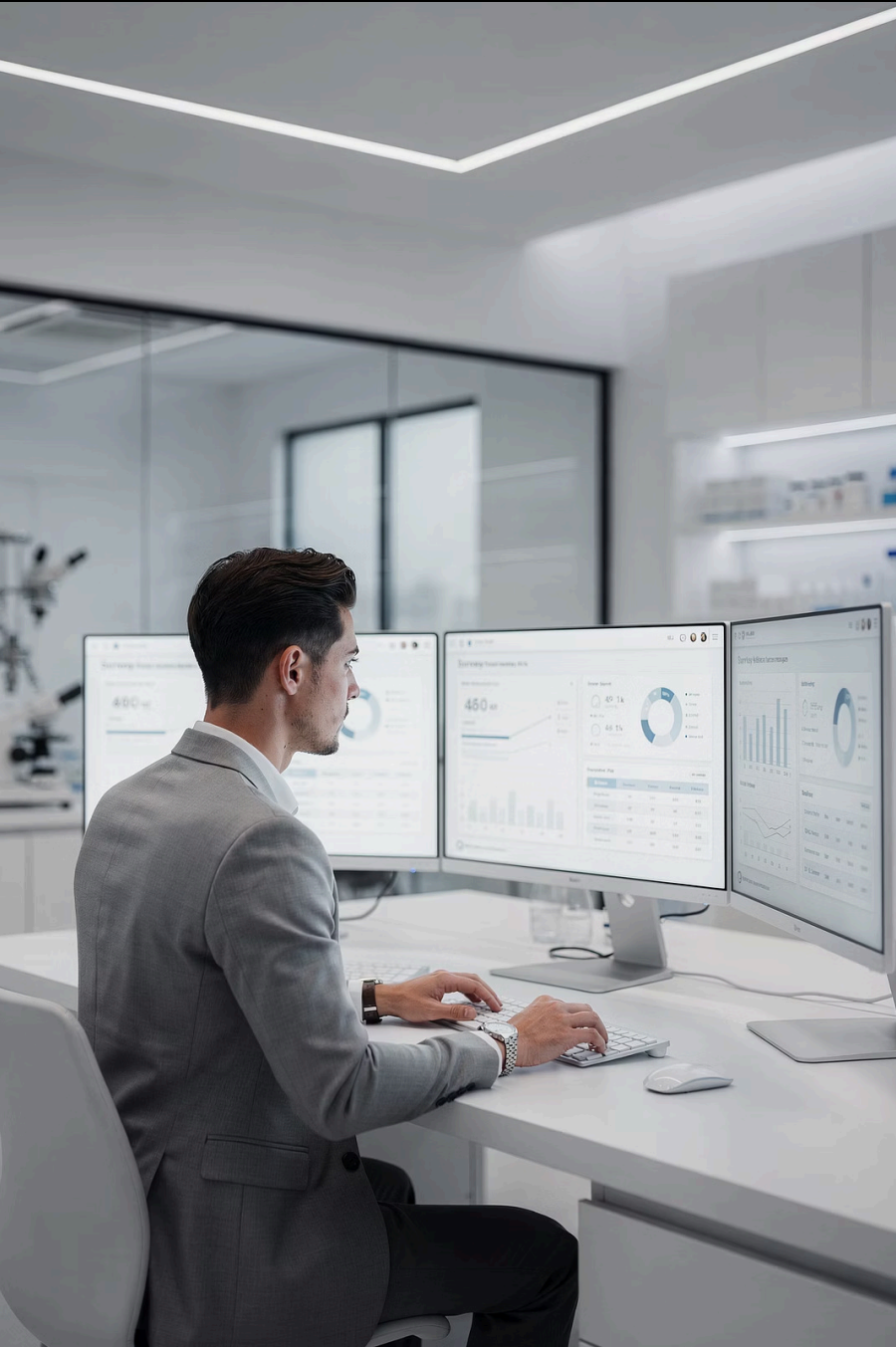
Secondary RQs (4)

Factorial validity (RQ2), experience and role moderation (RQ3), intention-usage relationship (RQ4), and user typology (RQ5)

03

12 Hypotheses Tested

7 core UTAUT2 paths (H1a–g), 1 AI Trust (H2), 2 moderation (H3–H4), 2 behavioral validation (H5–H6)

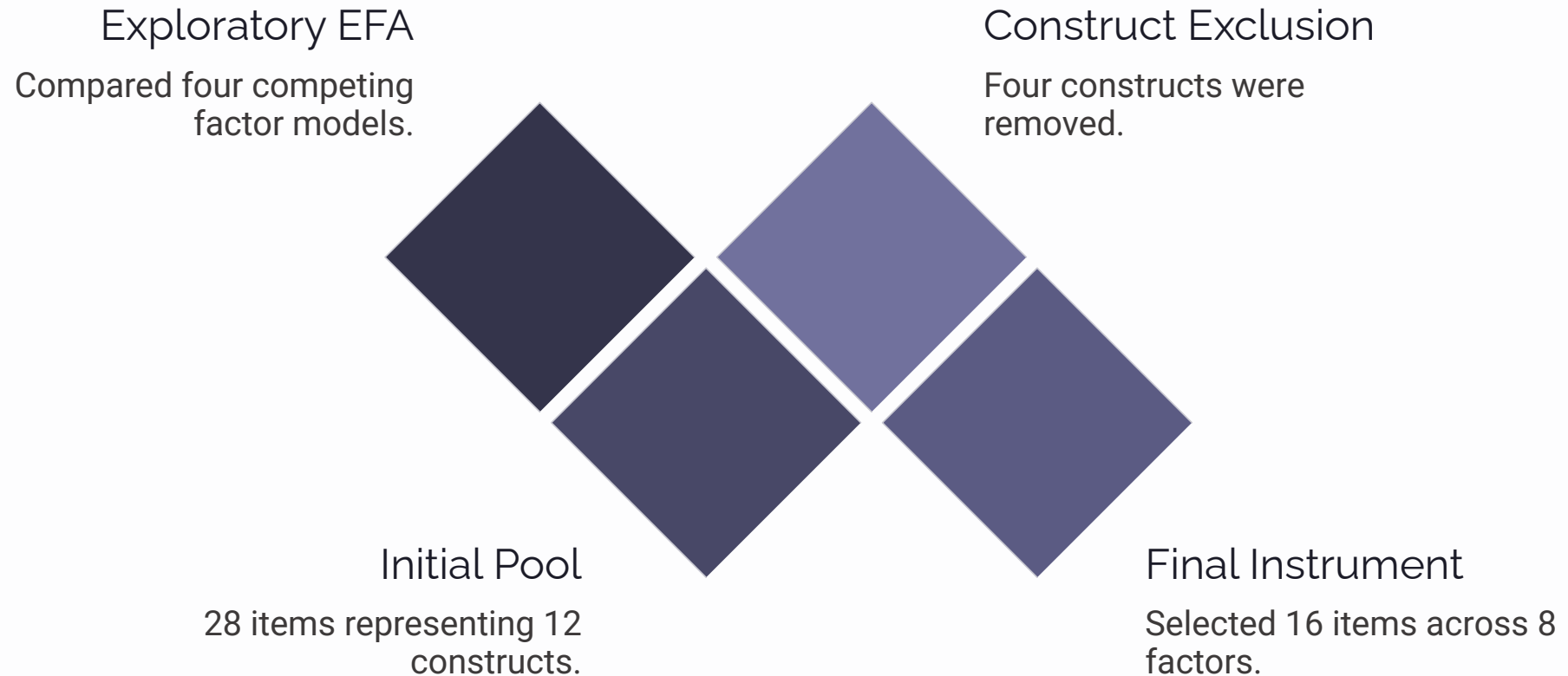


CHAPTER 2

Methodology

A post-positivist, sequential mixed-methods design with quantitative primacy. The quantitative phase (psychometric validation and structural equation modeling) provided the primary evidence base; qualitative content analysis offered contextual triangulation.

Instrument Development Process



Model D – the 8-factor, 16-item solution – was selected from four competing EFA models based on optimal factor structure, reliability, and theoretical interpretability.

Four Constructs Excluded

Voluntariness $\alpha = .41$

Items conflated organizational mandates with personal freedom

Explainability $\alpha = .58$

Items mixed understanding with preference

Ethical Risk $\alpha = .55$

Items confounded job displacement with data privacy

AI Anxiety $\alpha = .30$

Items collapsed avoidance and approach coping behaviors

Sample & Data Collection

Sample Overview

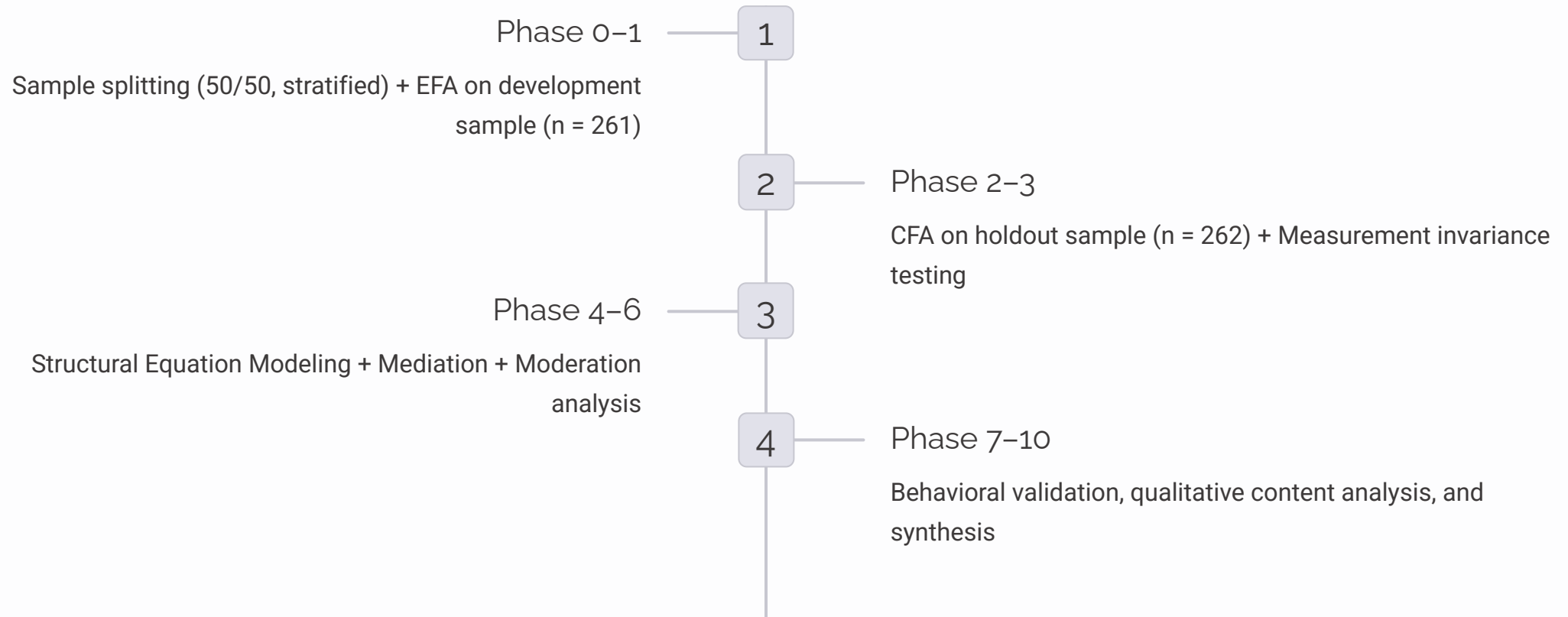
N = 523 U.S. adults recruited via Centiment using topic-blinded recruitment — respondents did not know the survey concerned AI, mitigating self-selection bias.

Collection period: October–November 2025 (3-week window)

Role Distribution

- **Academic** (students): n = 216 (41.3%)
- **Professional** (individual contributors, freelancers): n = 184 (35.2%)
- **Leader** (managers, executives): n = 123 (23.5%)
- **Disability status:** 68 participants (13.0%) reported a disability

10-Phase Analysis Pipeline



Dual-platform validation across Python and R/lavaan 0.6.21. CFA/SEM used Maximum Likelihood with Satorra-Bentler robust corrections (MLM). Bootstrap validation: 1,000 resamples.

CHAPTER 3

Key Findings: Measurement Model

The 16-item AIRS demonstrated excellent data adequacy ($KMO = .937$, Bartlett's $\chi^2 = 4,668.45$, $p < .001$) and strong confirmatory factor analysis results on the independent holdout sample.



CFA Model Fit: Excellent Performance

.975

CFI

Excellent (threshold $\geq .95$)

.960

TLI

Excellent (threshold $\geq .95$)

.065

RMSEA

Good (threshold $\leq .08$)

.026

SRMR

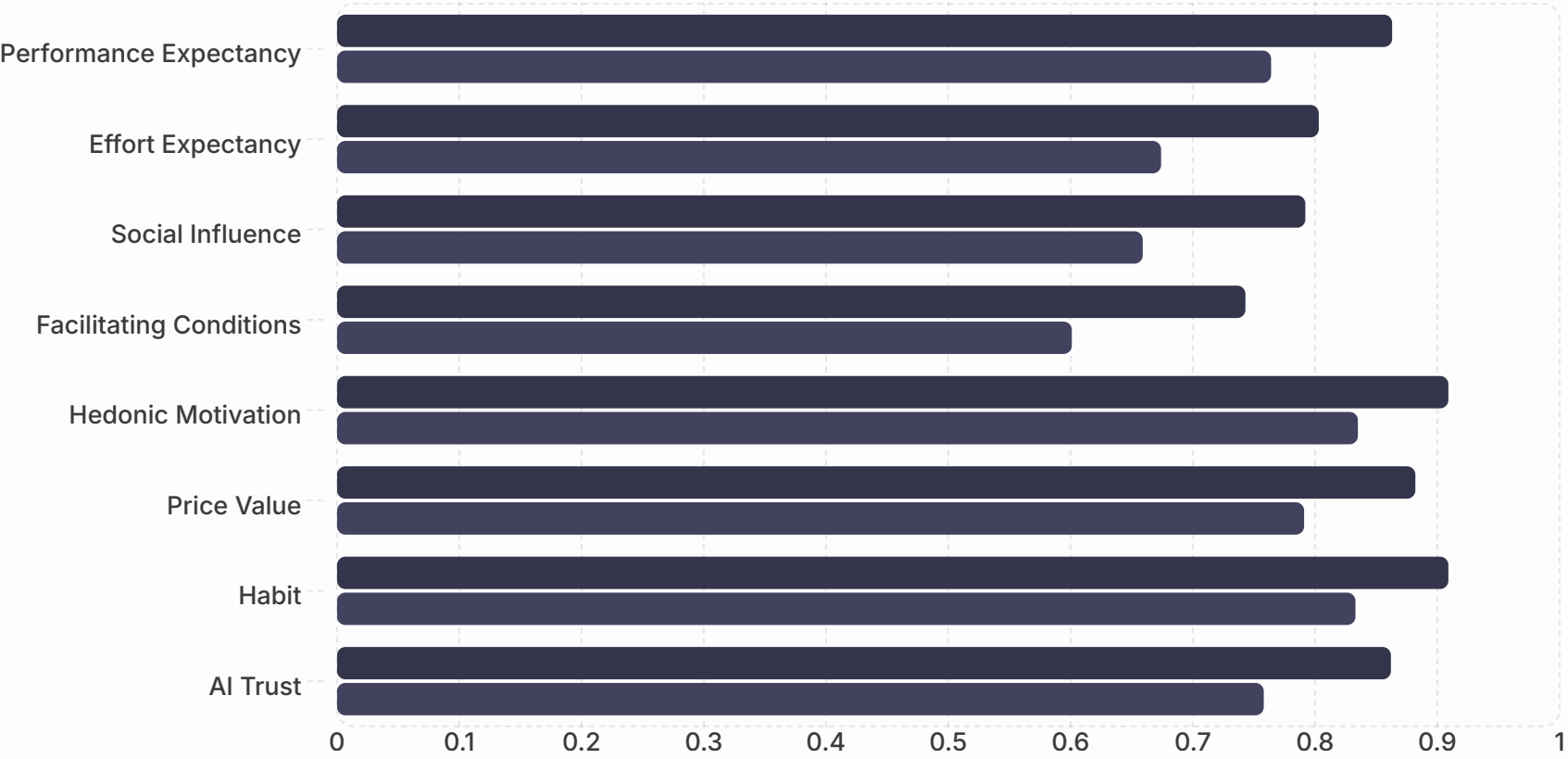
Excellent (threshold $\leq .08$)

Full-sample CFA confirmed these results: CFI = .979, TLI = .966, RMSEA = .061, SRMR = .022. The 8-factor model achieved excellent fit on the independent holdout sample (n = 262), confirming structural validity.

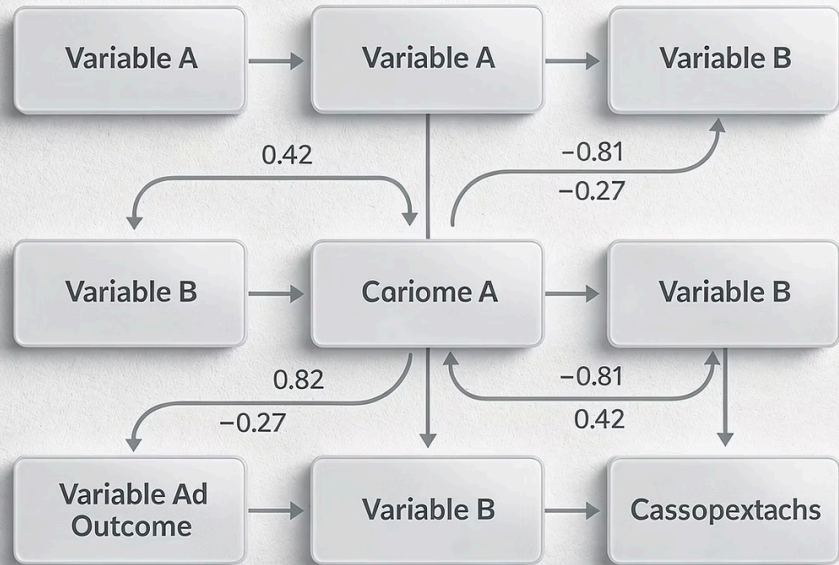
Reliability Across All 8 Factors

■ Cronbach's α ■ AVE

Factor



All composite reliabilities exceeded .750 and all AVE values exceeded .50, confirming convergent validity across every construct.



Structural Model: Hypothesis Results

89.7%

Variance Explained
R² in Behavioral Intention
– an exceptionally high
value

$\beta = .505$

Price Value
Strongest predictor,
confirmed robust via
bootstrap

5 of 12

Hypotheses
Supported
Plus 3 partial supports
and 4 not supported

Path Coefficients: Full Results

Hypothesis	β	p	Result
H1a: PE $\rightarrow$ BI	-.028	.791	Not Supported
H1b: EE $\rightarrow$ BI	-.008	.875	Not Supported
H1c: SI $\rightarrow$ BI	.136	.024	Supported (suggestive)
H1d: FC $\rightarrow$ BI	.059	.338	Not Supported
H1e: HM $\rightarrow$ BI	.217	.014	Supported (suggestive)
H1f: PV $\rightarrow$ BI	.505	<.001	Confirmed — Strongest
H1g: HB $\rightarrow$ BI	.023	.631	Not Supported
H2: TR $\rightarrow$ BI	.106	.064	Marginal

The Price Value Dominance Finding

Prior UTAUT Literature

Blut et al. (2022) meta-analysis across 417+ studies: **Performance Expectancy** was the dominant predictor (weighted $\beta = .39$)

Present Study

Price Value dominated ($\beta = .505$) while Performance Expectancy was non-significant ($\beta = -.028$).

Interpretation: When users already regard AI's usefulness as a baseline expectation, adoption decisions shift to whether perceived benefits justify the financial investment.

AI Trust: Diagnostic Retention Rationale

Statistical Picture

AI Trust did not reach conventional significance ($p = .064$). Model comparison favored the 7-factor UTAUT2-only model on parsimony grounds ($\Delta AIC = +2.01$).

Why It Was Retained

Trust provides unique diagnostic capability — identifying trust deficits other constructs cannot detect. The marginal effect ($\beta = .106$) suggests theoretical relevance; detecting it at 80% power requires $n > 600$ (present $N = 523$ yields ~68% power).

Organizational Value

Organizations gain actionable insight from separately measuring trust, enabling targeted confidence-building interventions that no other AIRS construct can prescribe.

Moderation Effects

Experience Moderation (H3: Partial)

Professional experience significantly moderated HM $\rightarrow$ BI ($\beta = .136$, $p = .007$). Experienced users weight enjoyment more heavily.

Low-usage group: Performance Expectancy mattered ($\beta = .371$)

High-usage group: Price Value dominated ($\beta = .878$)

Population Moderation (H4: Partial)

Hedonic Motivation showed significant population moderation ($p = .041$):

- **Academic group:** HM $\beta = +0.449$ (strong positive)
- **Professional group:** HM $\beta = -0.301$ (negative)

Academics adopt AI for exploration; professionals may view "enjoyment" as irrelevant to task-focused work.

Behavioral Validation

$$\rho = .69$$

BI $\rightarrow$ Usage

Spearman correlation between
Behavioral Intention and actual AI
tool usage ($p < .001$)

$$\rho = .57$$

ChatGPT

Individual tool correlation with
Behavioral Intention

$$\rho = .54$$

MS Copilot

Individual tool correlation with
Behavioral Intention

$$\rho = .52$$

Google Gemini

Individual tool correlation with
Behavioral Intention

AIRS measures a construct with real behavioral consequences. Role differences confirmed (H6): Leaders > Professionals > Academics across tool breadth, usage frequency, and intensity (all $p < .001$).

Three-Segment User Typology



AI Enthusiasts

31% (n = 162)

Early adopters with high engagement across all dimensions. Mean BI = 4.23. Deploy as change champions and peer influencers.



Moderate Users

47% (n = 246)

Pragmatic users near population means. Respond to value demonstrations and social proof from peers.



AI Skeptics

22% (n = 115)

Resistant users with below-mean readiness scores. Mean BI = 1.71. Require trust-building before engagement strategies will succeed.

The three-cluster solution explained **65.9% of BI variance** ($F(2,520) = 503.47, p < .001, \eta^2 = .659$), confirming a clear readiness gradient.

Disability & AI Anxiety: A Key Finding

Core Finding

Participants with disabilities (n = 68, 13%) reported significantly higher AI-related anxiety than non-disabled participants.

M = 2.89 vs. M = 2.48 · Cohen's d = 0.36 · p = .006

Broader Pattern

Participants with disabilities also reported significantly less enjoyment from AI tools (HM: d = .28, p = .030). All eight constructs showed consistent directional differences favoring the non-disabled group — a coherent disadvantage profile.



Qualitative Themes

Open-ended responses (n = 243, 46.5% response rate) revealed five major themes:



Positive Experience (24.7%)

Most prevalent theme – users describing satisfying AI interactions



Work & Productivity (19.3%)

Second most prevalent – efficiency and output gains



Human Element (13.6%)

Concerns about human judgment and collaboration



Learning & Education (13.6%)

Academics emphasized educational applications



Accuracy & Reliability (10.7%)

Trust-related concerns about AI outputs

CHAPTER 4

Theoretical Contributions

This study advances technology acceptance theory in six distinct ways — from the first empirical UTAUT2 validation for AI tools to novel mechanisms explaining the adoption-value gap.



Contributions 1–3: Theory Advances

1 First Empirical UTAUT2 Validation for AI

The 8-factor structure confirms UTAUT2's architecture applies to AI, but predictor importance shifts substantially. Four traditional predictors (PE, EE, FC, HB) were non-significant.

2 Price Value Dominance — AI as Distinct Category

PV ($\beta = .505$) displacing PE challenges decades of UTAUT research. In AI contexts, utility functions as a baseline expectation; adoption is driven by cost-benefit ratios.

3 Career Development × Technology Acceptance

Experience moderation ($p = .007$) and population moderation (Academic $\beta = +0.449$ vs. Professional $\beta = -0.301$) establish that adoption mechanisms are career-stage dependent.

Contributions 4–6: Applied Advances

1

Three-Segment User Typology

Empirically derived typology grounded in validated psychometric dimensions — explains 65.9% of BI variance, confirming validity as a practical classification tool.

2

Converging 2025 Industry Validation

McKinsey (N = 1,993), BCG, MIT Media Lab, Gartner, Deloitte, and others independently reported patterns consistent with AIRS findings — centrality of perceived value, heterogeneous readiness, limits of one-size-fits-all strategies.

3

Adoption-Value Gap Mechanisms

Four specific mechanisms identified: (1) value communication misalignment, (2) heterogeneous readiness, (3) neglected affective barriers, (4) context-inappropriate frameworks.

CHAPTER 5

Practical Implications

The AIRS findings translate directly into actionable guidance for organizations deploying AI, tool designers, trainers, and policymakers.



For Organizations: Segment-Targeted Rollout



Enthusiasts (31%)

Deploy as change champions and peer influencers. Provide access to advanced capabilities.



Moderate Users (47%)

Provide value demonstrations and social proof. Show concrete ROI and peer success stories.



Skeptics (22%)

Address trust barriers and anxiety first. Generic training will not move this group.

📌 **Key principle:** Frame AI adoption in value terms — time saved, errors reduced, revenue generated. Capability showcases alone are less effective than concrete cost-benefit demonstrations.

8 Structured Practitioner Recommendations

01

Diagnose Before Prescribing

Conduct baseline AIRS assessment before designing AI adoption initiatives

03

Attend to Affective Barriers

Hedonic motivation and anxiety influence adoption beyond rational calculations

01

Differentiate by Experience

Novice users focus on utility; experienced users focus on value

03

Design for Accessibility

Universal design reduces anxiety-driven non-adoption among users with disabilities

02

Lead With Value

Frame AI in cost-benefit terms (PV dominance), not capability showcases

04

Leverage Social Influence

Peer champions and leadership endorsement are the third significant predictor (SI $\beta = .136$)

02

Monitor Trust Continuously

Trust is a diagnostic indicator and necessary condition, not just a predictor

04

Consider Segment Heterogeneity

Any single intervention strategy will fail for at least one-third of the workforce

Limitations

Methodological

- **Cross-sectional design** — prevents causal inference about relationship direction or how intentions evolve over time
- **Self-reported intention** — BI-Usage correlation ($\rho = .69$) confirms association, but intention does not perfectly predict behavior
- **Panel sampling** — Centiment panel may not perfectly represent the broader U.S. population

Measurement

- **2 items per factor** — constrains reliability and elevates inter-factor correlations; HTMT violations in PE–HM–PV triad are partly structural
- **4 dropped constructs** — AI Anxiety in particular warrants item redesign given its disability association
- **Western sample** — all U.S.-based; cross-cultural generalizability is limited

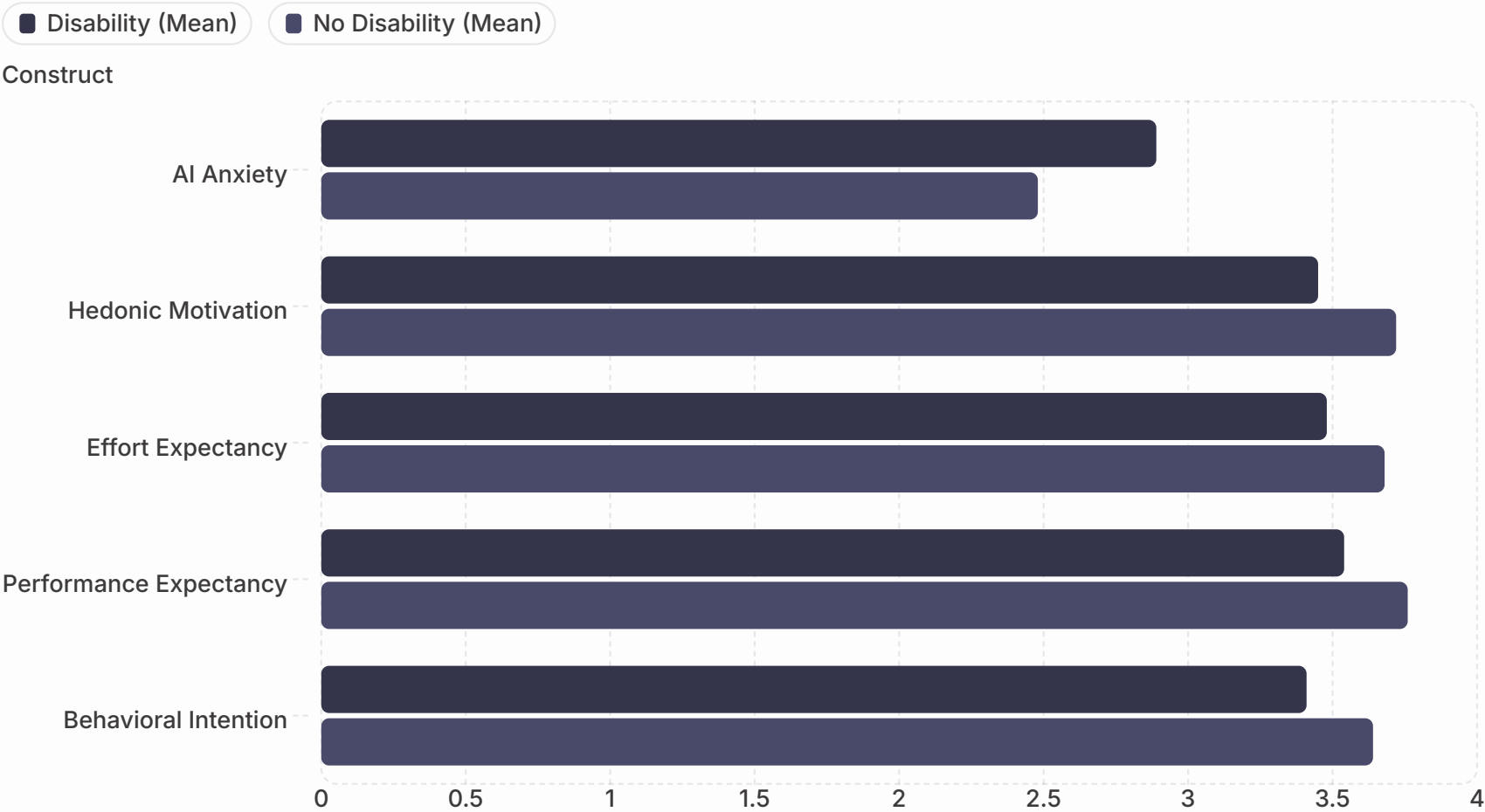
CHAPTER 6

Neurodiversity & Disability: Special Section

The AIRS dataset includes 68 participants (13.0%) who self-reported a disability — sufficient for exploratory between-group analysis. These findings are among the most practically salient in the dissertation.



The Disability Disadvantage Profile



Every positive adoption construct scores lower for participants with disabilities, while anxiety scores higher — a coherent disadvantage profile rather than random noise. AI Anxiety ($d = .36$, $p = .006$) and Hedonic Motivation ($d = .28$, $p = .030$) reached statistical significance.

Interpreting the Disability Findings

Anxiety Reflects Lived Experience

For users who have repeatedly encountered inaccessible interfaces or algorithmic bias, elevated AI anxiety is a rational response to historical exclusion — not a deficit to be corrected.

HM Suppression Is a Design Problem

The significant HM gap ($d = .28$) implies current AI tools are less enjoyable for users with disabilities — pointing to accessibility failures, not user deficits.

The Curb Cut Effect

Making AI tools more accessible would reduce anxiety-driven non-adoption broadly. Removing design-induced friction benefits the entire user population.

CHAPTER 7

Future Research Roadmap

A four-phase, multi-year research agenda designed to expand, validate, and operationalize the AIRS instrument across diverse populations and contexts.



Four-Phase Research Agenda



Phase 1: AIRS-28 Development (2026–2027)

Expand to 28 items (3–4 per factor). Redesign Explainability, Ethical Risk, and AI Anxiety items. Target: eliminate all HTMT violations.



Phase 3: Intervention Framework (2028–2029)

Design and test AIRS-informed organizational change programs. Confirm whether AIRS-guided interventions outperform generic AI training.



Phase 2: Longitudinal & Cross-Cultural (2026–2028)

12-month longitudinal panel tracking readiness evolution. Validate AIRS across non-Western cultural contexts.



Phase 4: Comprehensive Prediction System (2029+)

Integrate AIRS with behavioral data, organizational context, and industry-specific moderators for a full AI readiness monitoring system.

Appropriate Reliance: A New Research Direction

"The goal should not be maximum adoption but **calibrated adoption** — knowing when to use AI and when to rely on human judgment."

The AETHER group's Calibrated AI Reliance (CAIR) and Calibrated Self-Reliance (CSR) constructs reframe the research program from "more adoption is better" to "calibrated adoption is optimal." This line of inquiry is extended in *The Life of Alex Finch* (AlexBooks, March 2026), a documentary biography whose "Calibrated Confidence" chapter addresses over-reliance, verification, and preservation of human judgment.

2026 Contextual Developments

DeepSeek R1 Release

Open-source frontier AI capabilities are no longer exclusive to well-resourced organizations — fundamentally changing the Price Value calculation for all users.

Agentic AI Systems

AI operating with greater autonomy raises new trust and control questions beyond the original AIRS scope — new constructs will be needed.

EU AI Act Enforcement

Regulatory compliance emerges as a new adoption consideration absent from the current UTAUT2 framework — expanding the construct space required.

These developments do not invalidate the AIRS findings but indicate the construct space will need to expand to remain comprehensive.



CHAPTER 8

AIRS Enterprise: Research to Practice

The AIRS Enterprise platform (airs.correax.com) translates the validated AIRS-16 into a production SaaS system — delivering practical assessment tools to organizations while enabling longitudinal data collection at scale.

Platform Core Capabilities



5-Minute Assessment

Interactive AIRS-16 with progress tracking, save/resume, and smart follow-up questions



AIRS Score & Typology

Composite readiness score (8–40, $r = .876$ validity) with 94.5% accurate three-segment classification



Results Visualization

Animated gauge, 8-axis construct radar chart, and typology badge with actionable context in 29 languages



AI-Personalized Guides

Streaming LLM-generated action plans (3–5 pages) tailored to individual profile and career context via Azure OpenAI GPT-4o-mini

Three Intervention Frameworks

TRANSFORM

Skeptics (Score ≤ 20)

Trust-building through graduated autonomy
— low-risk tasks first, demonstrating reliability through small wins, progressively expanding scope with human-in-the-loop safeguards.

REDESIGN

Moderate Users (Score 21–30)

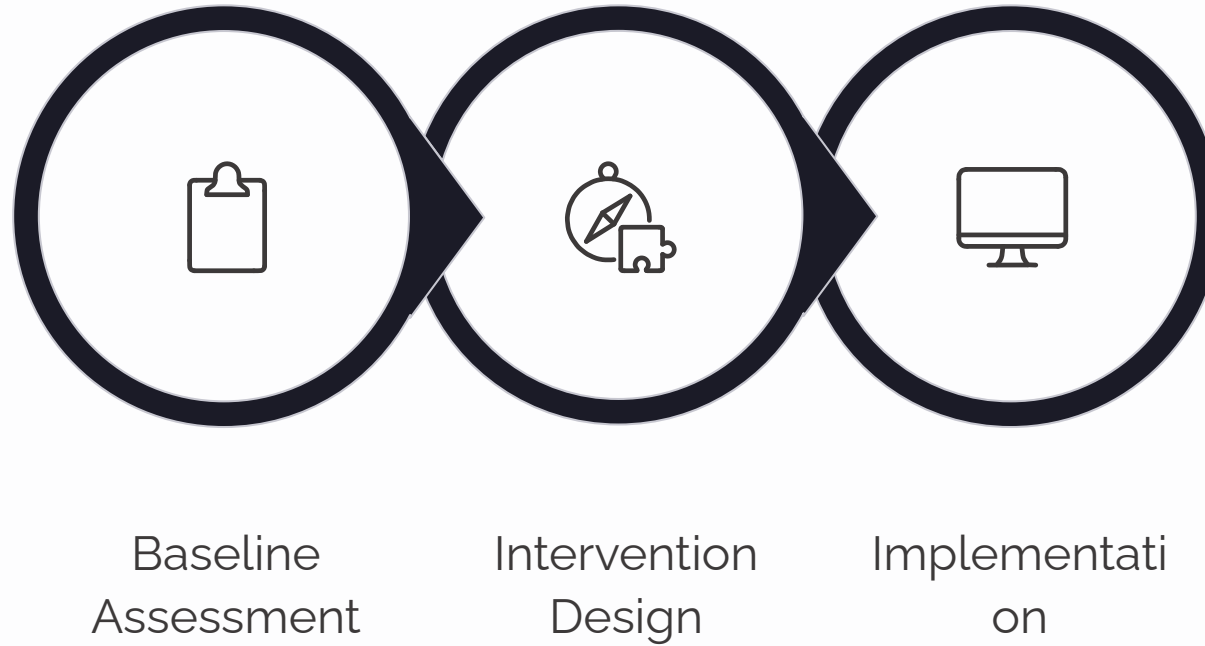
Workflow integration and ROI demonstration
— identifying specific processes where AI adds measurable value, social proof through peer success stories, structured skill development.

AMPLIFY

Enthusiasts (Score > 30)

Champion network activation — leveraging high-readiness individuals as change agents, providing advanced AI capabilities, creating mentorship structures that accelerate adoption.

Organizational Engagement Model



Every organizational deployment generates structured data feeding directly into the normative benchmarking database — building population-specific norms across 8+ target groups planned for 2026–2028.

The AIRS-16 Final Instrument

16 predictor items across 8 constructs + 4 Behavioral Intention outcome items, measured on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

Construct	Item 1	Item 2
PE	AI tools help me accomplish tasks more quickly.	Using AI improves the quality of my work or studies.
EE	Learning to use AI tools is easy for me.	Interacting with AI tools is clear and understandable.
SI	People whose opinions I value encourage me to use AI tools.	Leaders in my organization or school support the use of AI tools.
FC	I have access to training or tutorials for the AI tools I use.	The AI tools I use are compatible with other tools or systems I use.
HM	Using AI tools is stimulating and engaging.	AI tools make my work or studies more interesting.
PV	I get more value from AI tools than the effort they require.	Using AI tools is worth the learning curve.
HB	Using AI tools has become a habit for me.	I tend to rely on AI tools by default when I need help with tasks.
TR	I trust AI tools to provide reliable information.	I trust the AI tools that are available to me.



Conclusion: What AIRS Delivers

Psychometric Excellence

CFI = .975, all $\alpha > .74$, all CR > .75.
Explains 89.7% of variance in AI adoption intention.

Central Finding

Price Value ($\beta = .505$) — not
Performance Expectancy — is the
strongest predictor of AI adoption
intention.

Practical Tool

16-item instrument supports large-scale administration with 8-factor diagnostic detail for targeted intervention design.

The Research-Practice Bridge

Three Complementary Paths

The AIRS instrument, the Enterprise platform, and *The Life of Alex Finch* create complementary paths for advancing the research across academic, organizational, and public audiences.

Bidirectional Flow

Every organizational deployment strengthens the normative database. Every research finding improves the intervention recommendations delivered to practitioners. This bidirectional flow embodies the applied mission of the DBA: producing scholarship that directly improves organizational decision-making.

📄 **Degree:** Doctor of Business Administration (DBA) · **Sample:** N = 523 U.S. students and professionals · **Instrument:** AIRS-16 (8 factors × 2 items + 4 BI items) · **Platform:** airs.correax.com